

AIRLINE DATA STATISTICAL ANALYSIS

R Programming Project



Course Title: Programming with R

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Contents

1	Introduction	2
2	Setup and Data Import	2
3	New Variable Creation	2
4	Question 1: Coach Ticket Price Analysis	2
4.1	R Code	2
4.2	Interpretation	3
5	Question 2: 8-Hour Flight Analysis	3
5.1	R Code	3
5.2	Interpretation	4
6	Question 3: Flight Delay Distribution	4
6.1	R Code	4
6.2	Interpretation	4
7	Question 4: Correlation with Coach Price	4
7.1	R Code	4
7.2	Interpretation	5
8	Question 5: Coach vs First-Class Price	5
8.1	R Code	5
8.2	Interpretation	5
9	Question 6: WiFi, Meal and Entertainment Effect	5
9.1	R Code	5
9.2	Interpretation	6
10	Question 7: Passengers vs Flight Hours	6
10.1	R Code	6
10.2	Interpretation	6
11	Question 8: Weekend vs Weekday Price	6
11.1	R Code	6
11.2	Interpretation	7
12	Question 9: Redeye vs Non-Redeye Flights	7
12.1	R Code	7
12.2	Interpretation	7
13	Question 10: Full Statistical Analysis	7
13.1	R Code	7
13.2	Interpretation	8
14	Conclusion	8

1 Introduction

This project presents a complete statistical analysis of airline data using R Programming. Different statistical methods, visualization techniques, correlations, and regression analysis were performed to understand airline ticket pricing, delays, passenger behavior, and other flight-related factors.

2 Setup and Data Import

```
1 #wd Setup
2 getwd()
3 setwd("C:/Users/Ashraful/Downloads/New_folder")
4 getwd()
5 list.files()
6
7 airline_random_1000 <- read.csv(
8 "C:/Users/Ashraful/Downloads/New_folder/airline_random_1000.csv")
9
10 head(airline_random_1000)
11 View(airline_random_1000)
12 str(airline_random_1000)
```

3 New Variable Creation

```
1 my_id <- 10004
2
3 airline_random_1000$new_miles <-
4 airline_random_1000$miles + my_id
5
6 airline_random_1000$new_passengers <-
7 airline_random_1000$passengers + my_id
8
9 airline_random_1000$new_coach_price <-
10 airline_random_1000$coach_price + my_id
11
12 airline_random_1000$new_firstclass_price <-
13 airline_random_1000$firstclass_price + my_id
14
15 View(airline_random_1000)
```

4 Question 1: Coach Ticket Price Analysis

4.1 R Code

```
1 mean(airline_random_1000$new_coach_price)
2
3 max(airline_random_1000$new_coach_price)
4
5 min(airline_random_1000$new_coach_price)
6
7 median(airline_random_1000$new_coach_price)
8
9 sd(airline_random_1000$new_coach_price)
10
11 hist(airline_random_1000$new_coach_price,
12       main = "Histogram of Coach Ticket Price",
13       xlab = "Coach Ticket Price",
14       ylab = "Frequency",
15       col = "skyblue",
16       border = "black")
17
18 boxplot(airline_random_1000$new_coach_price,
19         main = "Boxplot of Coach Ticket Price",
20         ylab = "Coach Ticket Price",
21         col = "red",
22         border = "black")
```

4.2 Interpretation

- Mean shows the average coach ticket price.
- Maximum and minimum indicate highest and lowest prices.
- Standard deviation explains variability.
- Histogram shows data distribution.
- Boxplot identifies outliers.

5 Question 2: 8-Hour Flight Analysis

5.1 R Code

```
1 flight_8_hr <- subset(airline_random_1000, hours == 8)
2
3 summary(flight_8_hr$new_coach_price)
4
5 mean(flight_8_hr$new_coach_price)
6
7 min(flight_8_hr$new_coach_price)
8
9 max(flight_8_hr$new_coach_price)
```

5.2 Interpretation

- Analysis focuses on only 8-hour flights.
- Average, minimum, and maximum prices are calculated.
- Helps understand long-distance flight pricing.

6 Question 3: Flight Delay Distribution

6.1 R Code

```
1 summary(airline_random_1000$delay)
2
3 hist(airline_random_1000$delay,
4       main = "Flight_Delay_Distribution",
5       xlab = "Delay_Minutes",
6       ylab = "Frequency",
7       col = "orange",
8       border = "black")
9
10 boxplot(airline_random_1000$delay,
11          main = "Delay_Boxplot",
12          ylab = "Delay_Minutes",
13          col = "yellow")
```

6.2 Interpretation

- Most flights delay between 0–20 minutes.
- Some flights experience large delays.
- Boxplot identifies extreme delays.

7 Question 4: Correlation with Coach Price

7.1 R Code

```
1 cor(airline_random_1000$coach_price,
2      airline_random_1000$new_miles)
3
4 cor(airline_random_1000$coach_price,
5      airline_random_1000$new_passengers)
6
7 cor(airline_random_1000$coach_price,
8      airline_random_1000$delay)
9
10 cor(airline_random_1000$coach_price,
11      airline_random_1000$hours)
```

```

12
13 plot(airline_random_1000$new_miles,
14       airline_random_1000$new_coach_price,
15       main = "New_Miles_vs_New_Coach_Price",
16       xlab = "Miles",
17       ylab = "Coach_Price",
18       col = "blue")

```

7.2 Interpretation

- Longer distance flights cost more.
- Flight hours positively affect prices.
- Delay has weak correlation with price.

8 Question 5: Coach vs First-Class Price

8.1 R Code

```

1 cor(airline_random_1000$new_coach_price,
2      airline_random_1000$new_firstclass_price)
3
4 plot(airline_random_1000$new_coach_price,
5       airline_random_1000$new_firstclass_price,
6       main = "Coach_vs_First_Class_Price",
7       xlab = "Coach_Price",
8       ylab = "First_Class_Price",
9       col = "darkgreen")

```

8.2 Interpretation

- Strong positive correlation exists.
- Higher coach prices mean higher first-class prices.

9 Question 6: WiFi, Meal and Entertainment Effect

9.1 R Code

```

1 aggregate(new_coach_price ~ inflight_wifi,
2           data = airline_random_1000,
3           mean)
4
5 aggregate(new_coach_price ~ inflight_meal,
6           data = airline_random_1000,
7           mean)

```

```

8
9 aggregate(new_coach_price ~ inflight_entertainment,
10           data = airline_random_1000,
11           mean)

```

9.2 Interpretation

- Flights with WiFi cost more.
- Meals and entertainment increase prices.
- Extra facilities improve travel experience.

10 Question 7: Passengers vs Flight Hours

10.1 R Code

```

1 cor(airline_random_1000$passengers,
2     airline_random_1000$hours)
3
4 plot(airline_random_1000$hours,
5      airline_random_1000$passengers,
6      main = "Hours vs Passengers",
7      xlab = "Flight Hours",
8      ylab = "Passengers",
9      col = "purple")

```

10.2 Interpretation

- Weak positive relationship exists.
- Longer flights tend to carry more passengers.

11 Question 8: Weekend vs Weekday Price

11.1 R Code

```

1 aggregate(new_coach_price ~ weekend,
2           data = airline_random_1000,
3           mean)
4
5 boxplot(new_coach_price ~ weekend,
6         data = airline_random_1000,
7         main = "Weekend vs Weekday Prices",
8         col = "cyan")
9
10 t.test(new_coach_price ~ weekend,
11        data = airline_random_1000)

```

11.2 Interpretation

- Weekend flights are slightly more expensive.
- T-test checks statistical significance.
- p-value below 0.05 means significant difference.

12 Question 9: Redeye vs Non-Redeye Flights

12.1 R Code

```

1 aggregate(new_coach_price ~ redevye,
2           data = airline_random_1000,
3           mean)
4
5 boxplot(new_coach_price ~ redevye,
6         data = airline_random_1000,
7         main = "Redeye vs Non-Redeye",
8         col = "pink")

```

12.2 Interpretation

- Redeye flights are generally cheaper.
- Overnight flights are less preferred.

13 Question 10: Full Statistical Analysis

13.1 R Code

```

1 summary(airline_random_1000)
2
3 mean(airline_random_1000$new_coach_price)
4
5 median(airline_random_1000$new_coach_price)
6
7 sd(airline_random_1000$new_coach_price)
8
9 cor(airline_random_1000[,c("coach_price",
10                          "miles",
11                          "delay",
12                          "hours",
13                          "passengers")])
14
15 model <- lm(coach_price ~ hours +
16            miles +
17            delay,
18            data = airline_random_1000)

```

```
19  
20 summary(model)
```

13.2 Interpretation

- Miles significantly increase ticket price.
- Flight hours positively affect prices.
- Delay has smaller impact.
- Regression explains variable relationships.

14 Conclusion

This project successfully analyzed airline ticket pricing, delays, passenger counts, flight duration, and airline facilities using R Programming. Different statistical techniques such as descriptive statistics, correlation analysis, hypothesis testing, visualization, and regression analysis were used to understand the airline dataset effectively.

THANK YOU